GridIQ

Site assessment: overview

Demo Corporation · Headquarters

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment
GQ-A-0002 · 30-day assessment

Report
revision 0 · provisional preview

Observation period (SAST)
2026-10-22 21:00 -> 2026-11-11 21:11

Observed
20.01 days of 30

Site address
123 Power Lane, Johannesburg

Measurement points
Phase1 Test Meter (incomer), Main Incomer (incomer)

1. Executive findings

- Investigate (Phase1 Test Meter, confidence high; for: GridIQ): Only 0 % of the expected measurements were received; conclusions for this point are limited.
- Investigate (Main Incomer, confidence high; for: GridIQ): Only 65 % of the expected measurements were received; conclusions for this point are limited.

2. Installation and measurement boundary

Point	Role	Nominal	CT primary	Contracted MDI	Device · installed (SAST)
Phase1 Test Meter	incomer	230 V	–	300 kVA	device · 2026-11-03 20:52
Main Incomer	incomer	230 V	–	300 kVA	device · 2026-07-26 22:50

Point	Coverage	Samples	Reporting interval	Gaps >= 30 min	Meter
Phase1 Test Meter	0.0 %	3	32 s	2	device
Main Incomer	64.6 %	3 722	300 s	1	device

Coverage = samples received × the meter's median reporting interval ÷ the observation window. Missing data is never filled or shown as zero.

sum of 2 top-level incomer(s); sub-meters are inside these and are not added

3. Energy, demand and operating pattern

Phase1 Test Meter

Energy imported
1,7 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
–

Contracted MDI
300 kVA (–)

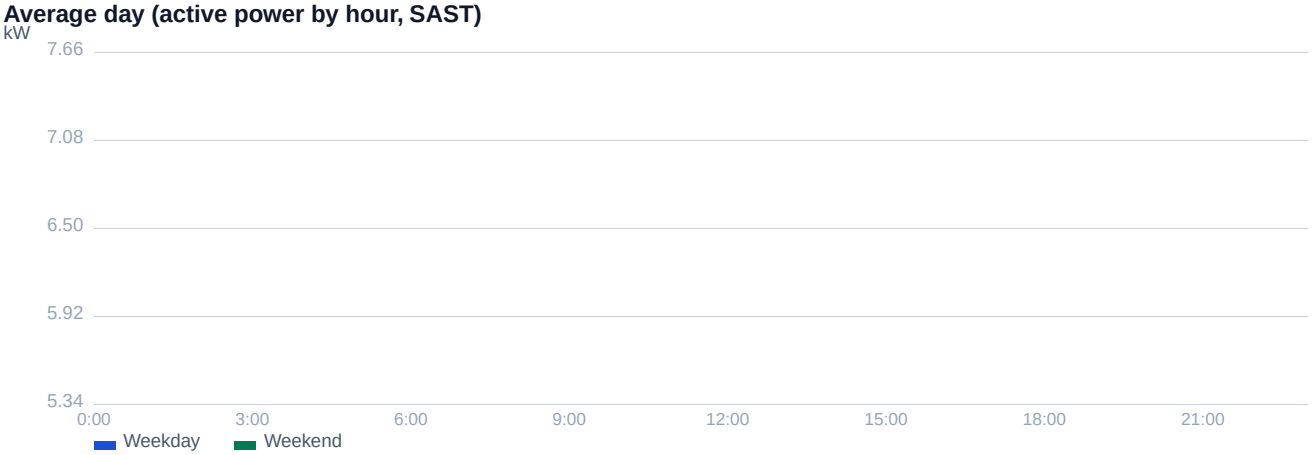
Highest instantaneous active power
7,2 kW (minute of 2026-11-03 20:53)

Baseload (10th percentile)
6,5 kW

Operating-hours average
–

Outside operating hours
6,5 kW

Average day (active power by hour, SAST)



Daily maximum active power (SAST days)

hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

Main Incomer

Energy imported
24 430,8 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
190,5 kVA at 2026-11-03 14:00

Contracted MDI
300 kVA (64 % used)

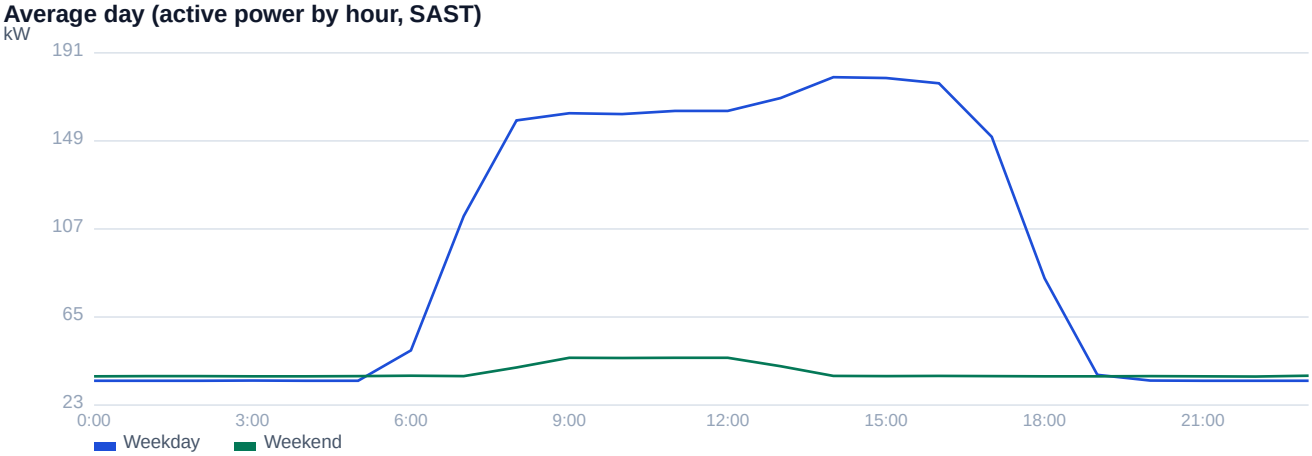
Highest instantaneous active power
192,4 kW (minute of 2026-11-03 14:00)

Baseload (10th percentile)
34,8 kW

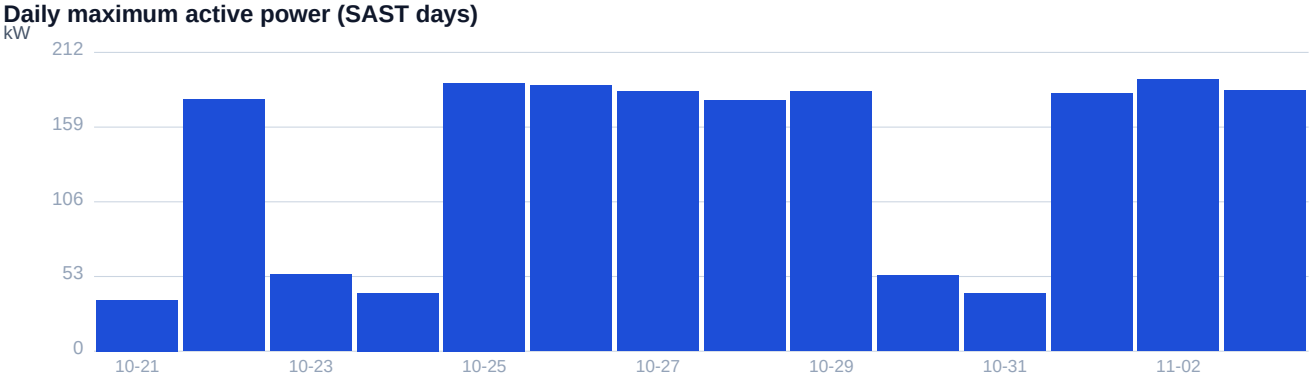
Operating-hours average
152 kW

Outside operating hours
39,1 kW

Average day (active power by hour, SAST)



Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

4. Electrical summary

- Phase1 Test Meter: voltage A 229,9–231,5 V, B 229,8–229,8 V, C 231–231 V; THD-V p95 not measured; PF (loaded) 0.957; 0 minute(s) below 23 V.
- Main Incomer: voltage A 226–234,7 V, B 225,7–233,9 V, C 226,6–235,6 V; THD-V p95 A 2.7 %; PF (loaded) 0.968; 0 minute(s) below 23 V.

Screening against NRS 048-2 thresholds; see the power-quality report for detail and the basis of every figure.

5. Notable events and data limitations

- PROVISIONAL: the assessment is still running; these results cover the elapsed period only and are not final.
- Observation period 20 days (2026-10-22T19:00:36.207Z to 2026-11-11T19:11:37.513Z); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.
- Phase1 Test Meter: 0 % of expected measurements received (2 gap(s) of 30 min or more).
- Phase1 Test Meter: the meter's capability profile is not confirmed; only channels present in the data are reported.
- Main Incomer: 65 % of expected measurements received (1 gap(s) of 30 min or more).
- Main Incomer: the meter's capability profile is not confirmed; only channels present in the data are reported.

0 voltage/frequency excursion event(s) recorded; inferred from the sampled RMS trend (pq_events), not device-recorded.

6. Prioritised actions

#	Action	Who investigates	Urgency	Confidence
1	Only 0 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high
2	Only 65 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high

7. Feasibility-data readiness

Supplied: 30-minute interval load profile (kW, kVA); daily energy; billed maximum demand with timestamps; power-quality screening results; metering topology and observation limits.

Missing inputs: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

GridIQ supplies measured data; system sizing and financial modelling belong to the feasibility study.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
86b4e86c0aedf5f1a9bcc50905be6438…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-11 21:11

Generated (SAST)
2026-11-11 21:11

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 86b4e86c0aedf5f1a9bcc50905be643808a12b576f4d88f9e6f9640329b8d249